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Self Healing Concrete: An Innovative Solution To Structural Cracking

Under the Guidance

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Contents

Introduction

- Self-healing concrete is a type of advanced concrete that can repair cracks and damages autonomously.
- Extends structure lifespan and reduces maintenance.
- It contains embedded healing agents, such as :-
  a) Bacteria that produce calcite.
  b) Microcapsules with healing compounds.
  c) Shape-memory alloys.
- When cracks occur, these agents are activated, releasing substances that seal the cracks and restore the concrete's integrity.
- In 2006, self-healing concrete was invented by microbiologist Professor Henk Jonkers at Delft University of Technology, Netherlands. This SHC, also known as self-repairing concrete, mimics the automatic appear, these capsules or fibers rupture, releasing the healing solution to seal the crack.

Problem Statement

- Define the Problem
  a. Concrete structures develop cracks due to loading, shrinkage, and environmental effects.
  b. These cracks reduce durability and allow harmful substances to enter the structure.
- Why it matters
  a. Cracks lead to corrosion of reinforcement and structural damage. Repair and maintenance increase the overall lifecycle cost.
  b. Durability is important for safe and sustainable infrastructure.
- Current Gap
  a. Conventional concrete cannot repair cracks automatically.
  b. Existing repair methods are costly and temporary.
  c. Need for smart materials like self-healing concrete.

Objective

- The objective of self-healing concrete is to develop concrete that can automatically repair its own cracks, improve durability and service life, prevent the ingress of harmful substances, and reduce maintenance and repair requirements without external intervention. Such as:-

  - Crack Repair Without External Intervention.
  - Enhancement of Durability.
  - Increase in Service Life of Structures.
  - Reduction in Maintenance and Repair Costs.
  - Improvement in Structural Safety and Reliability.
  - Sustainability and Environmental Benefits.
  - Enhancement of Mechanical and Durability Properties.

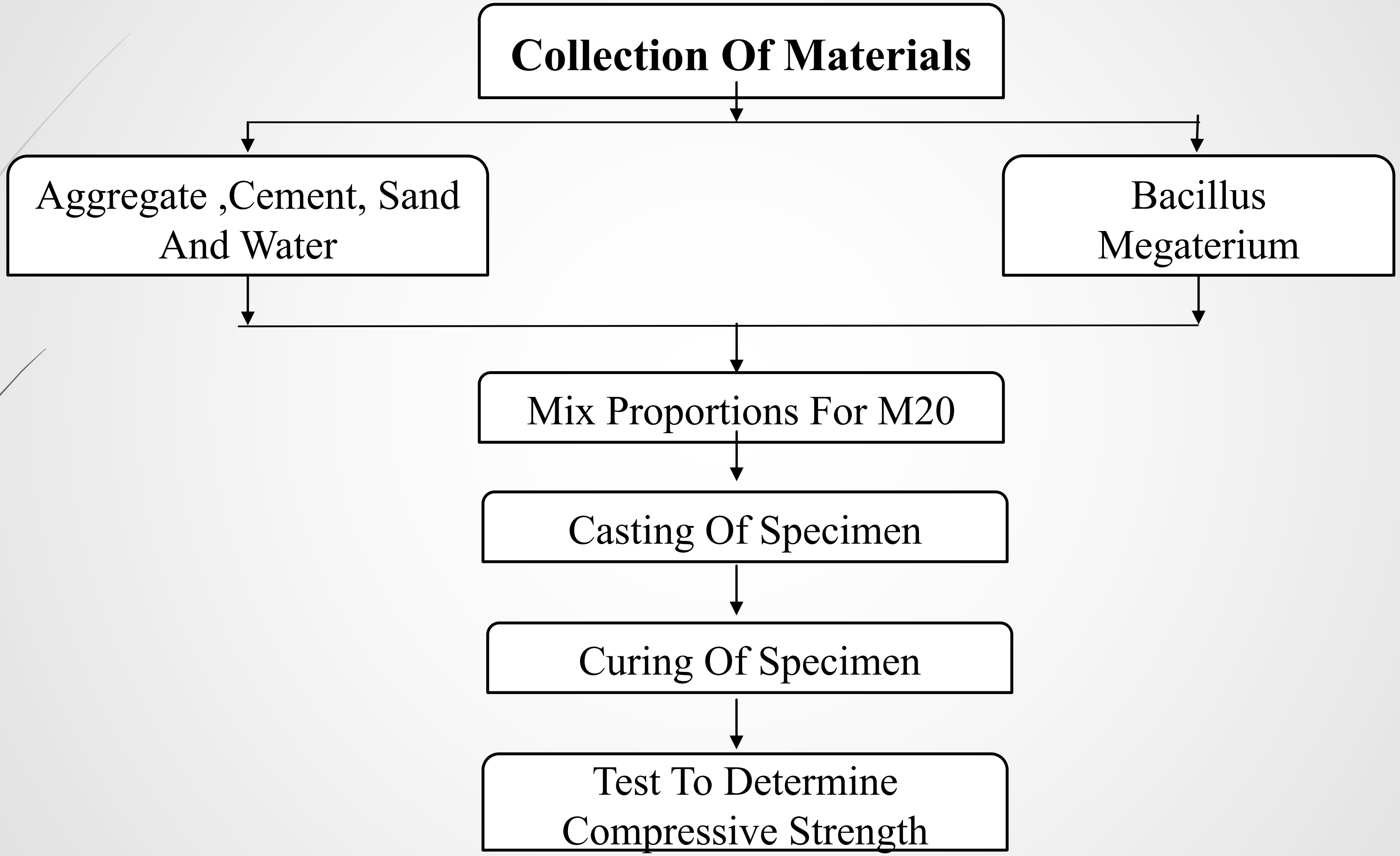
Literature Review

S.No.	Publisher	Year	Author	Result/Findings
1.	Construction and Building Materials 146 (2017) 419–428	2014	Gupta Souradeep Pang Sze Dai , Kua Harn Wei a	i. The initial cost of incorporating bacteria is quite high. ii. If infrastructures are built with self-healing concrete that is designed to function under multiple damages, very minimal cost may be accumulated over the lifetime.
2.	Construction Building Material 87 (2015) 1–7.	2014	M. Luo, C.x. Qian, R.-y. Li	i. It was observed that under immersion in water condition crack width up to 0.3 mm were completely healed and healing ratio of cracks between 0.1 and 0.3 mm were about 85%. ii. Specimens with bacterial spores displayed superior healing to control samples but its effectiveness was less than that of external healing.
3.	International journal of Recent Scientific Research.	2015	Rishabh Lala, Aslam Hussain, Salim Akhtar	i. While most healing agents are chemically based, more recently the possible application of bacteria as self-healing agent has also been considered. ii. Metabolically active bacteria consume oxygen; the healing agent may act as an oxygen diffusion barrier protecting the steel reinforcement against corrosion.

Methodology (Overview)

- Self-healing concrete is an advanced material designed to repair its own cracks automatically, increasing durability and reducing maintenance costs. The methodology mainly depends on how the healing mechanism is activated — either autogenic (natural) or autonomous (engineered).Approach
- Let's go step-by-step:-
  a. Grade: M20.
  b. Mix Ratio: 1 : 1.5 : 3.
  c. W/C Ratio: 0.50.
  d. Healing Agent: Bacteria (Bacillus subtilis) or microcapsules (1–3% cement).
  e. Mixing: Dry mix → add water with agent → mix well.
  f. Casting: Pour in moulds, compact.
  g. Curing: 28 days in water.
  h. Cracking: Make small cracks (0.1–0.5 mm).
  i. Healing: Keep in moist condition 7–28 days.
  j. Testing: Check crack closure & strength recovery.

Methodology (Flowchart)



Implementation

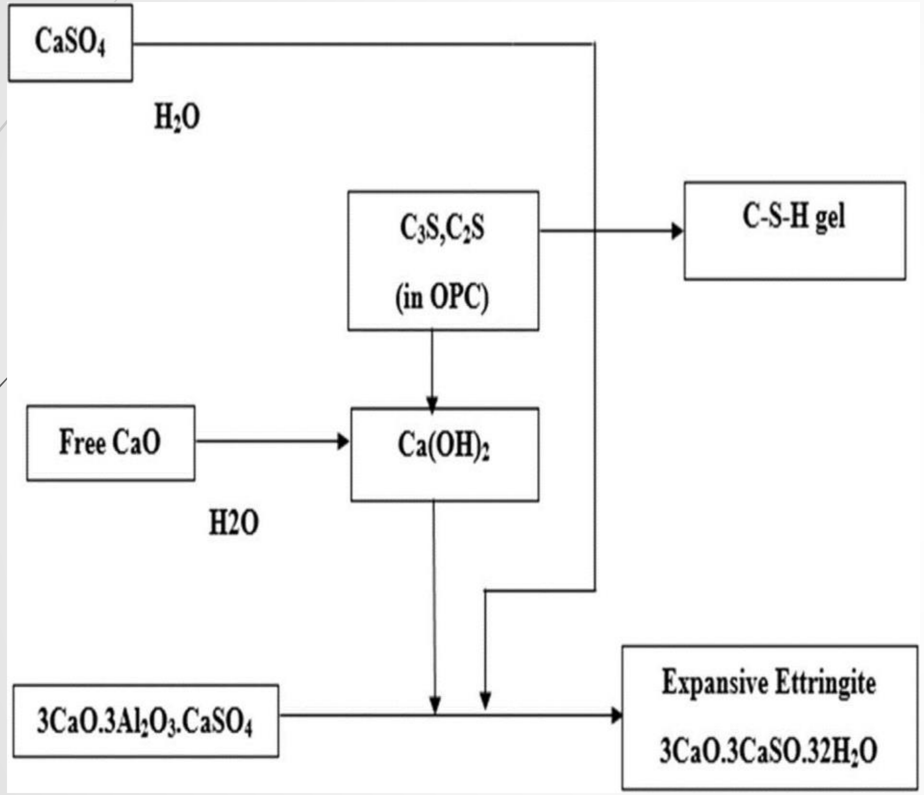


Fig.: Working mechanism and formation of healing products incorporating Calcium sulphate aluminate.

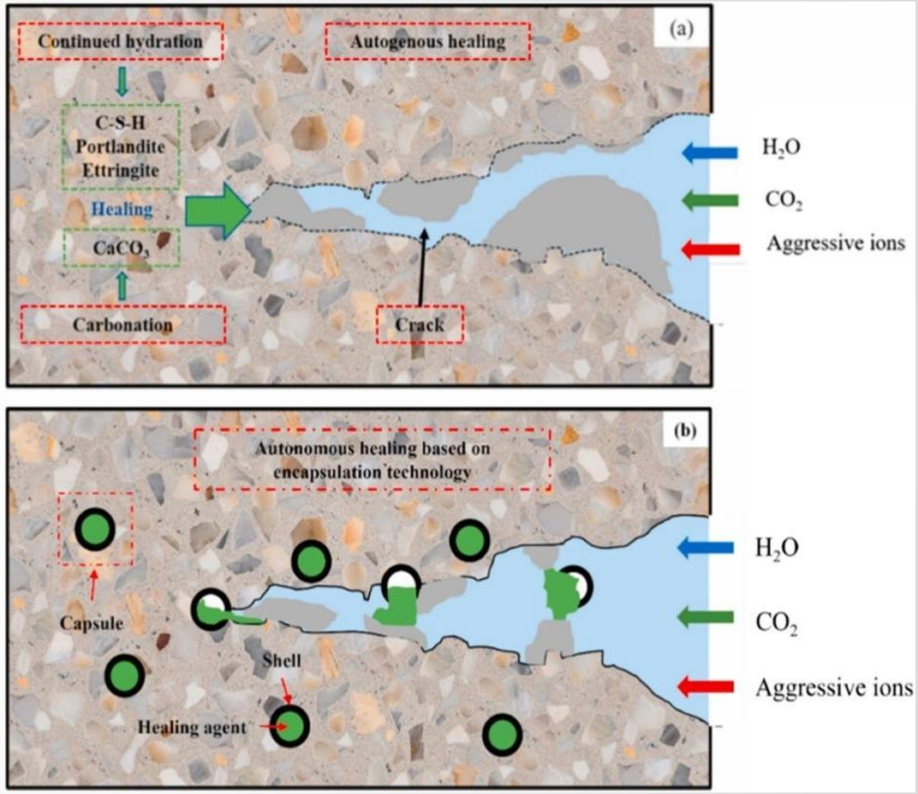


Fig.: Working mechanism of autogenous and autonomic self-healing in cementitious materials.

Proposed Methodology / Solution

- **Crack creation:** Create cracks in the specimen after compressive strength testing.
- **Healing process :** Allow the cracks to heal under controlled conditions.
- **Compressive strength:** Calculate the maximum load at failure.
- **Re-testing:** Re-test the healed specimen to evaluate the recovery of compressive strength.
- **Healing efficiency:** Calculate the percentage recovery of compressive strength after healing.
- **Failure mode:** Analyze the failure pattern to understand the self-healing mechanism.

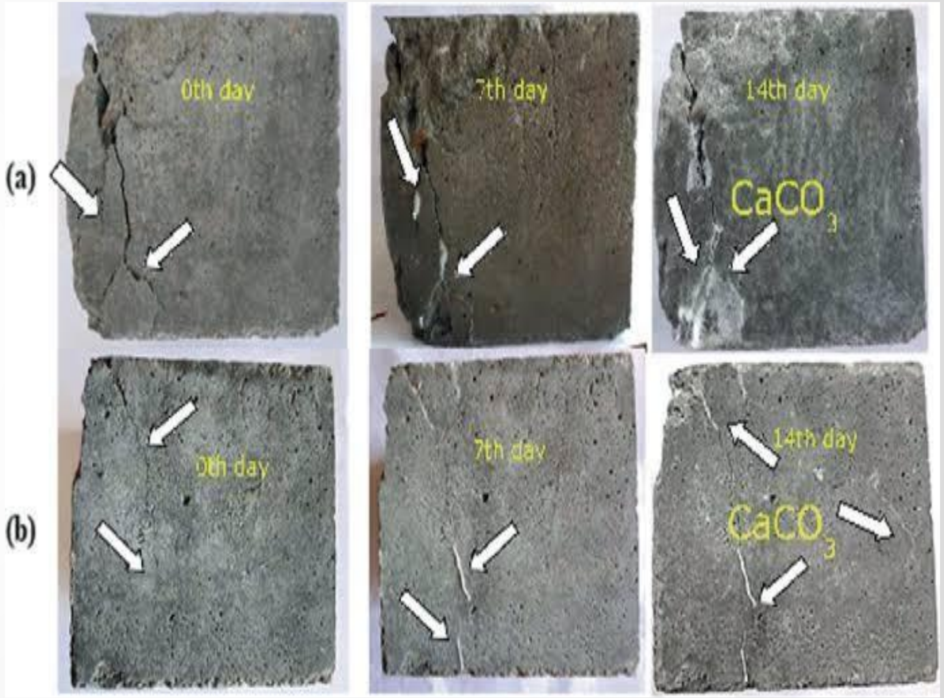
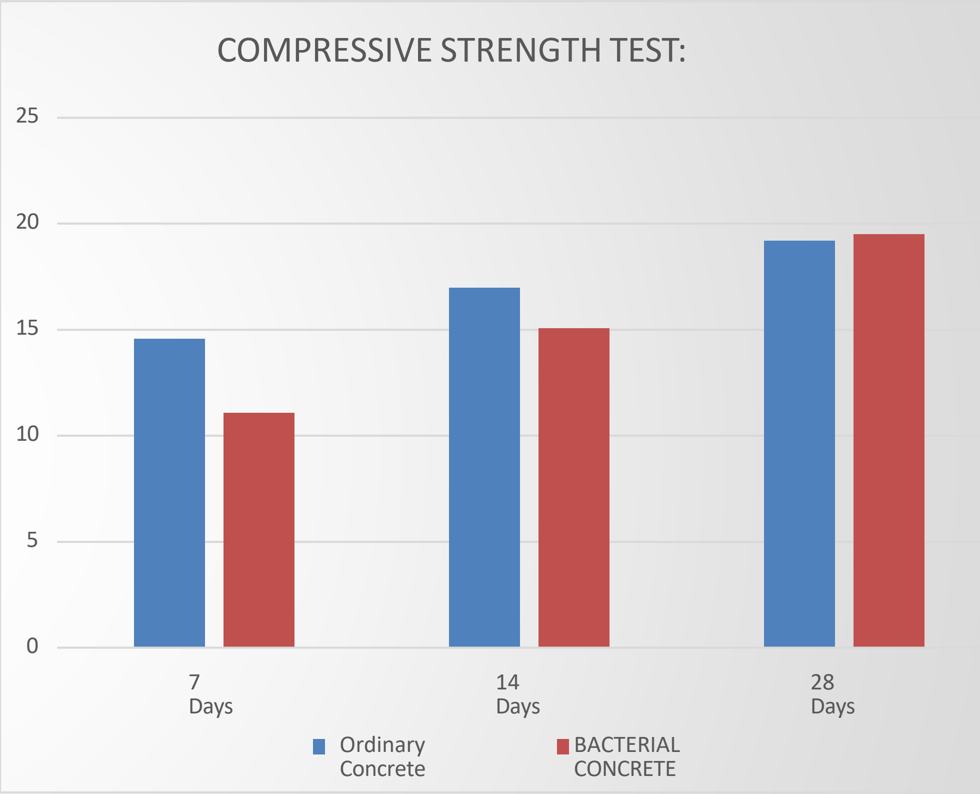


Fig.: Self-Healing Process Of Bacterial Concrete

Results

- ❖ In this research, the compressive strength of concrete cubes measuring 150 mm x 150 mm x 150 mm was tested in accordance with British Standard BS EN 12390-3:2019.
- ❖ After curing for 28 days, the cubes were removed from the curing tank, weighed, and then tested using a Compressive Testing Machine (CTM) at a loading rate of 19 MPa until failure. Each specimen was placed centrally in the machine, and the failure load was recorded.
- ❖ The compressive strength was calculated by dividing the recorded load by the cross-sectional area of each specimen.
- ❖ A total of 12 cubes of different samples were subjected to compression testing.
- ❖ We have found that bacteria improve concrete properties, increasing strength by 11.07 MPa in 7 days, 15.07 MPa in 14 days, and 19.50 MPa in 28 days.



Comparison of compressive strength of ordinary and bacterial M20

Analysis

Interpretation:

i. The results indicate that bacterial action helps in crack sealing through calcite formation.
ii. Long-term benefits outweigh the higher initial investment.
iii. The technology is suitable for critical and durable structures.

Comparison:

Sr. No.	Feature	Traditional Concrete	Self Healing Concrete
1.	Detection	Manual(Visual Inspection)	Autonomous(Sensors/Agents)
2.	Response Time	Delayed(Months/Years)	Immediate(Days/Weeks)
3.	Permeability	Increases as cracks grow	Decreases as cracks seal
4.	Service Life	Standard(30-50 Years)	Extended(80+ Years)
5.	Long-term Cost	High(Cumulative Repairs)	Low(Single investment)

Discussion

- What worked
  1. Self-healing mechanism successfully sealed micro-cracks in concrete.
  2. Bacterial calcite precipitation effectively filled the cracks.
  3. Reduction in water permeability was observed after healing.
  4. Strength and durability properties were maintained or slightly improved.
  5. Autonomous healing reduced the need for external repair.
- Limitations
  1. High initial cost.
  2. Effective mainly for small cracks.
  3. Requires moisture for healing.
  4. Bacteria survival is sensitive to environment.
  5. Limited large-scale practical use.

Conclusion

- The paper discusses methods for designing self-healing concrete, emphasizing the introduction of bacteria to enhance its properties beyond conventional concrete.
- It reviews various types of bacteria suitable for remedying cracks in concrete.
- Bacteria repair concrete cracks by producing calcium carbonate crystals, effectively blocking and repairing the cracks.
- We have found that bacteria improve concrete properties, increasing strength by 11.07% in 7 days, 15.07% in 14 days, and 19.50% in 28 days.
- The study examines different methods and techniques for designing self-healing concrete, categorizing them into chemical, biological, and natural self-healing processes. Chemical techniques have traditionally been used as the primary method for self-healing concrete.
- Repairing and maintaining concrete structures using conventional methods is costly compared to self-healing concrete. Therefore, it's essential to improve and implement these methods for better concrete structure durability.

Future Scope

- Self-healing concrete is a rapidly evolving technology, and researchers are working on several future enhancements to improve its performance, sustainability, and cost effectiveness.

- Advanced Self-Healing Mechanisms:-
  a. Shape Memory Alloys (SMAs)
  b. Microbial Self-Healing
  c. Nanomaterials
- Improved Durability and Sustainability:-
  a. Fiber-Reinforced Self-Healing Concrete
  b. Recycled Materials
  c. Bio-Based Self-Healing Concrete
- Smart Self-Healing Concrete:-
  a. Sensors and Monitoring Systems
  b. Artificial Intelligence (AI) and Machine Learning (ML)
  c. Self-Healing Concrete with Energy Harvesting

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Thank You

- Thank You!
- Any Questions ?